Logistic Regression

November 16, 2022

As you come in...

Complete the worksheet... we will reconvene in 20 mins...

Looking Ahead:

1. Today: Logistic Regression
2. Tuesday (November 15): In-class Regression Lab- Due November 29 if not completed
   - **Checkpoint 3 Due 11:59PM: Preliminary Analysis**
3. Thursday (November 17): No class!

Predictions with Multiple Regression

TABLE 4. The Effect of Being in the Rockets' Range on Right-Wing Parties' Vote-Shares

	DV: Right-Wing Vote-Share							
	(1)		(2)		(3)		(4)	
	Coef.	Std. err.	Coef.	Std. err.	Coef.	Std. err.	Coef.	Std. err.
In range	0.04***	(0.01)	0.06***	(0.01)	0.02***	(0.01)	0.04***	(0.01)
Suicide terror fatalities			0.02*	(0.01)	0.01	(0.01)	0.02***	(0.01)
Mean wage			0.10***	(0.01)	−0.01	(0.01)	−0.00	(0.01)
Population			0.02	(0.02)	−0.05**	(0.02)	−0.00	(0.00)
Median age			−0.00	(0.00)	−0.01***	(0.00)	0.00	(0.00)
Male-to-female ratio			−0.06	(0.06)	−0.04	(0.07)	0.09	(0.07)
Net migration			−0.08	(0.09)	−0.03	(0.08)	−0.01	(0.02)
Jews share			0.10	(0.07)	0.06	(0.04)	0.10***	(0.02)
High school graduates			−0.08***	(0.02)	−0.00	(0.04)	−0.10**	(0.04)
Origin from Asia in 1995							−0.04	(0.04)
Origin from Africa in 1995							0.03	(0.03)
Origin from the Soviet Union in 1995							0.05	(0.03)
Regional capital							0.02**	(0.01)
Distance to border							−0.00	(0.00)
Lagged DV							0.84***	(0.02)
Constant	0.40***	(0.00)	0.29	(0.24)	1.03***	(0.20)	−0.04	(0.09)
Locality fe	yes		yes		yes		no	
Year dummies	no		no		yes		no	
R^2	0.015		0.140		0.532		0.856	
No. of obs (localities)	1000 (269)		790 (257)		790 (257)		705 (254)	

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Predictions with Multiple Regression

Using Model 3: compute the predicted value for a locality that is <u>in range</u>, assuming all other variables are at their means.

$$\hat{Y}_i = 0.06(InRange) + 0.02(S.Terror) + 0.1(Wage) + 0.02(Population)...$$

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Using Model 3: compute the predicted value for a locality that is <u>in range</u>, assuming all other variables are at their means.

$$\hat{Y}_i = 0.06(InRange) + 0.02(S.Terror) + 0.1(Wage) + 0.02(Population)...$$

$$\hat{Y}_i = 0.06(1) + 0.02(10.93) + 0.1(15.98) + 0.02(7.12)...$$

Hehe... this is (kindof) easier in R...

Step 1: Estimate Model

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```
model1<-lm(VoteShare~InRange+SuicideTerror,
data=terrorismdataset)
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Step 2: predict() function

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predict(model1, data.frame(InRange=1, SuicideTerror=10.93))
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predict(model1, data.frame(InRange=1, SuicideTerror=10.93))
predict(model1, data.frame(InRange=0, SuicideTerror=10.93))
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Step 1: Estimate Model

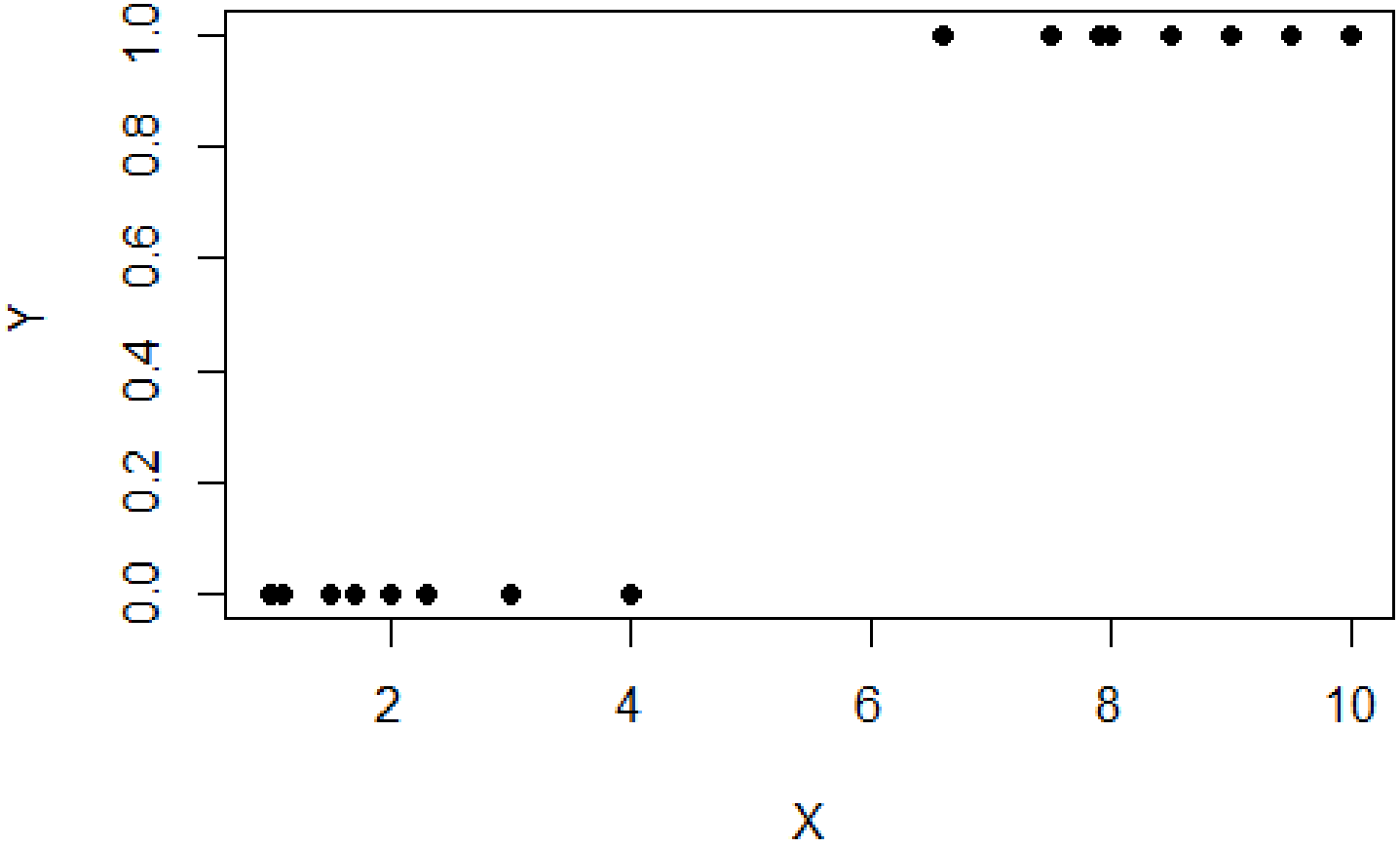
model1< − lm(VoteShare ∼ InRange + SuicideTerror, data=terrorismdataset)

Step 2: predict() function

predict(model1, data.frame(InRange=1, SuicideTerror=10.93))
predict(model1, data.frame(InRange=0, SuicideTerror=10.93))

When using predict(), must define variables exactly how they are spelled in model

Okay but what if...



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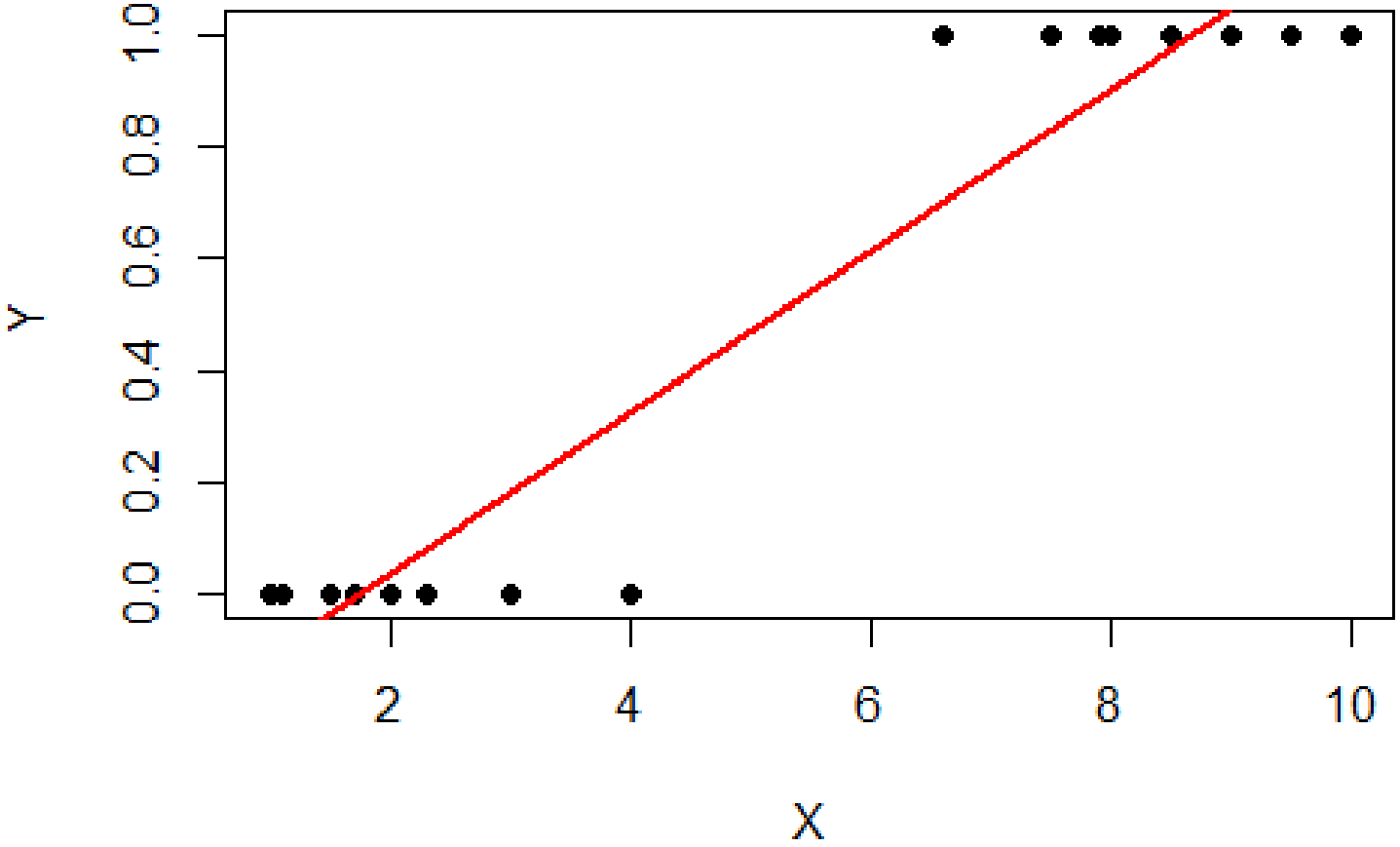
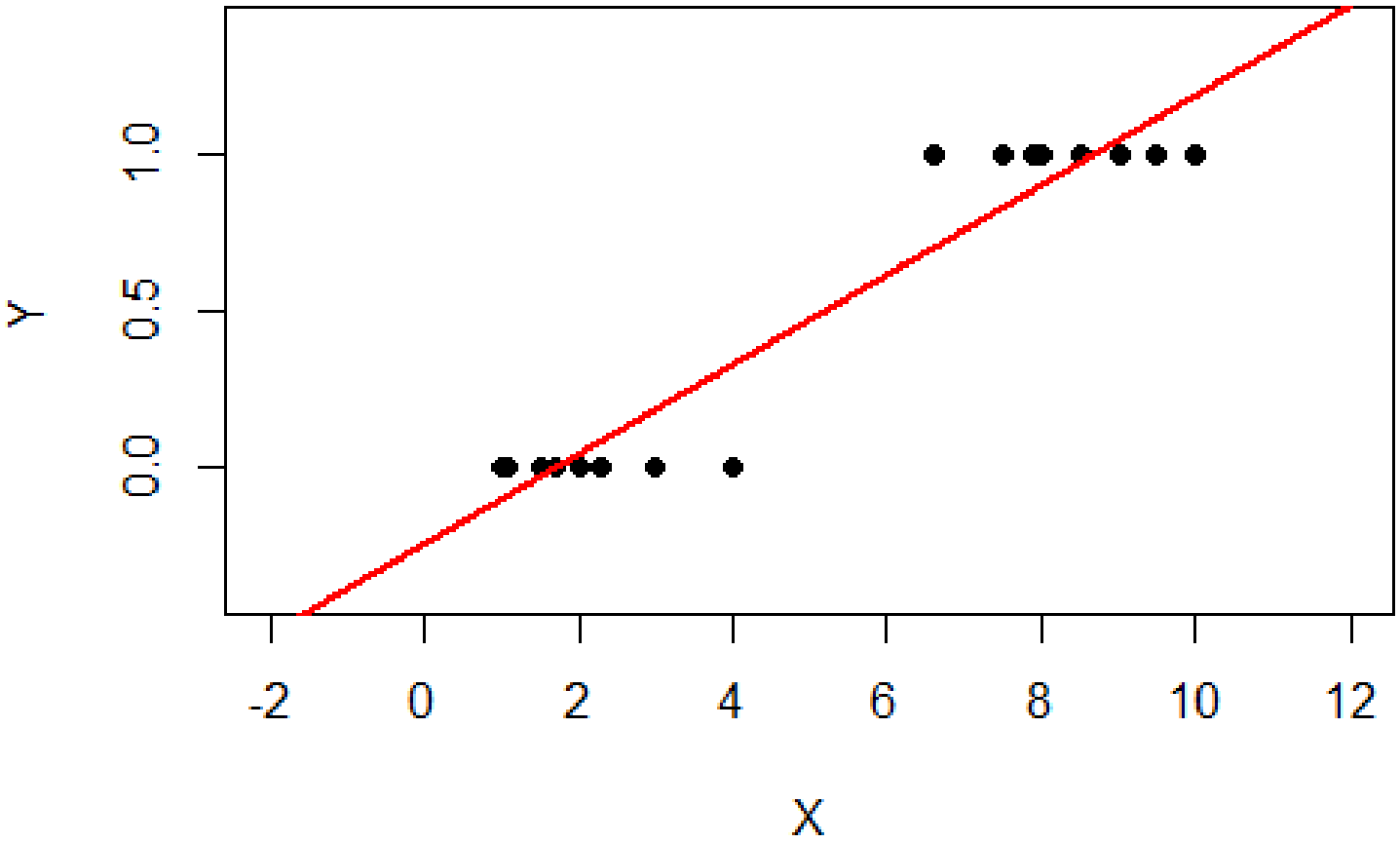


Linear Probability Model



Logistic Regression: Quick Math

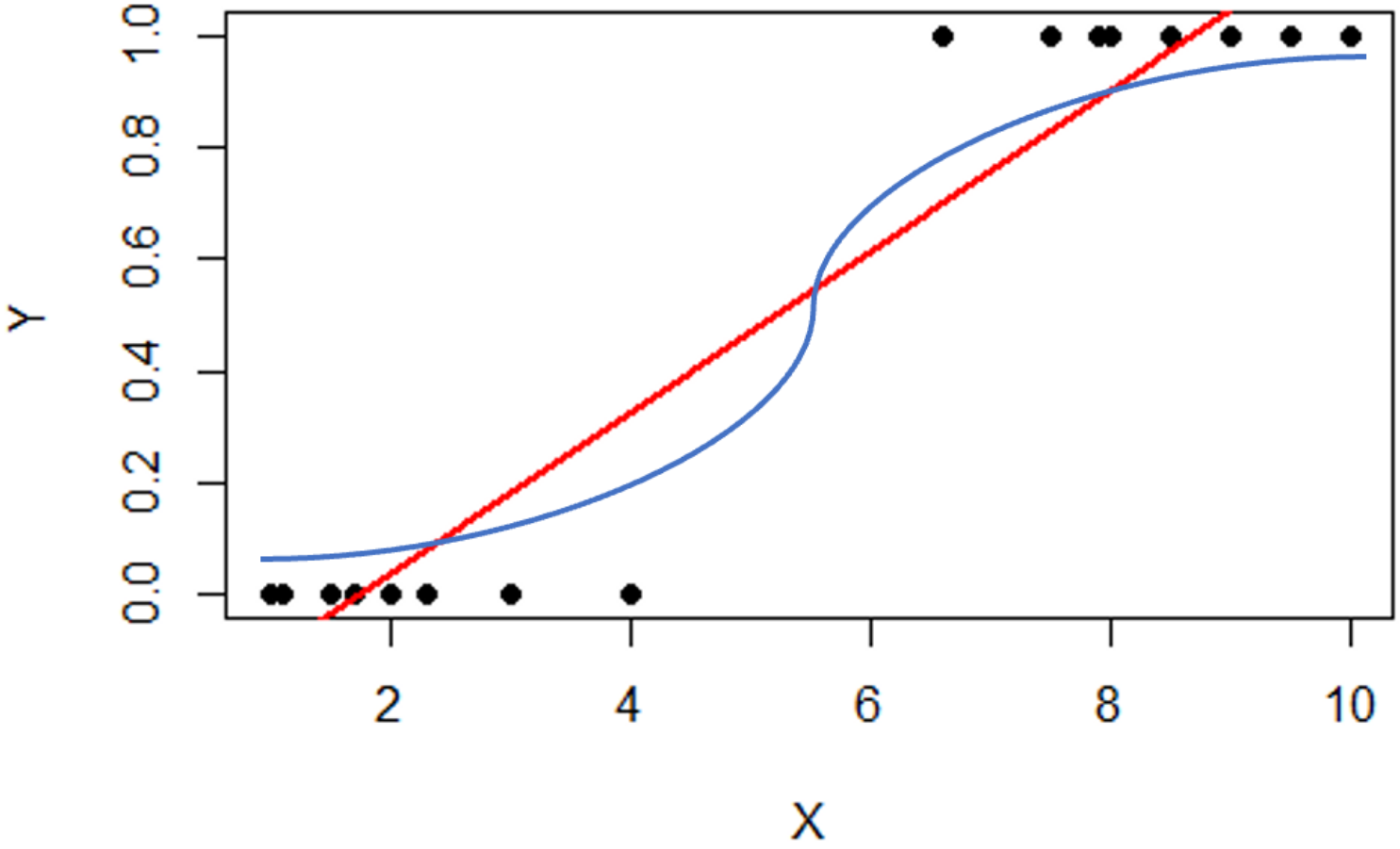
Instead of:

$$Y = \alpha + \beta_j X_j \tag{1}$$

Probabilities from logistic regression calculated as

$$Pr(Y = 1) = \frac{exp(\alpha + \beta_j)}{1 + exp(\alpha + \beta_j)} \tag{2}$$

Logistic Regression



Logistic Regression Conceptually

We use logistic regression when the Dependent Variable is binary.

- Benefits:
  1. "Bends" our mathy stuff to model the *probability of a 1 on the DV*
  2. Predictions will be realistic, given the values of the DV
- Downsides:

Logistic Regression Conceptually

We use logistic regression when the Dependent Variable is binary.

- Benefits:
  1. "Bends" our mathy stuff to model the *probability of a 1 on the DV*
  2. Predictions will be realistic, given the values of the DV
- Downsides:
  1. Cannot interpret coefficients in the same way...
     - Direction and significance is still the same interpretation
     - But magnitude of coefficients is not comparable
     - And doesn't just mean "a one unit increase in X is associated with a β increase in Y, on average"...

Logistic Regression Table

TABLE 1. Logit Analyses of Determinants of Civil War Onset, 1945–99

	Model				
	(1) Civil War	(2) "Ethnic" War	(3) Civil War	(4) Civil War (Plus Empires)	(5) Civil War (COW)
Prior war	−0.954**	−0.849*	−0.916**	−0.688**	−0.551
	(0.314)	(0.388)	(0.312)	(0.264)	(0.374)
Per capita income[a,b]	−0.344***	−0.379***	−0.318***	−0.305***	−0.309***
	(0.072)	(0.100)	(0.071)	(0.063)	(0.079)
log(population)[a,b]	0.263***	0.389***	0.272***	0.267***	0.223**
	(0.073)	(0.110)	(0.074)	(0.069)	(0.079)
log(% mountainous)	0.219**	0.120	0.199*	0.192*	0.418***
	(0.085)	(0.106)	(0.085)	(0.082)	(0.103)
Noncontiguous state	0.443	0.481	0.426	0.798**	−0.171
	(0.274)	(0.398)	(0.272)	(0.241)	(0.328)
Oil exporter	0.858**	0.809*	0.751**	0.548*	1.269***
	(0.279)	(0.352)	(0.278)	(0.262)	(0.297)
New state	1.709***	1.777***	1.658***	1.523***	1.147**
	(0.339)	(0.415)	(0.342)	(0.332)	(0.413)
Instability[a]	0.618**	0.385	0.513*	0.548*	0.584*
	(0.235)	(0.316)	(0.242)	(0.225)	(0.268)
Democracy[a,c]	0.021	0.013			
	(0.017)	(0.022)			
Ethnic fractionalization	0.166	0.146	0.164	0.490	−0.119
	(0.373)	(0.584)	(0.368)	(0.345)	(0.396)
Religious fractionalization	0.285	1.533*	0.326		1.176*
	(0.509)	(0.724)	(0.506)		(0.563)
Anocracy[a]			0.521*		0.597*
			(0.237)		(0.261)
Democracy[a,d]			0.127		0.219
			(0.304)		(0.354)
Constant	−6.731***	−8.450***	−7.019***	−6.801***	−7.503***
	(0.736)	(1.092)	(0.751)	(0.681)	(0.854)
N	6327	5186	6327	6360	5378

Note: The dependent variable is coded "1" for country years in which a civil war began and "0" in all others. Standard errors are in parentheses. Estimations performed using Stata 7.0. $^{*}p < .05$; $^{**}p < .01$; $^{***}p < .001$.

[a] Lagged one year.

[b] In 1000's.

[c] Polity IV; varies from −10 to 10.

[d] Dichotomous.

Logistic Regression in R

Generically:

```
model<-glm(Y~X1+X2, data=data, family="binomial")
```

Predicted Probabilities

$$Pr(Y = 1) = \frac{exp(\alpha + \beta_j)}{1 + exp(\alpha + \beta_j)} \tag{3}$$

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Gross.

Predicted Probabilities in R (Yay)

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```
model2<-glm(CivilWar~Mountains+PriorWar+GDPPC,
data=FL2003, family="binomial")
```

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model2< − glm(CivilWar ∼ Mountains + PriorWar + GDPPC, data=FL2003, family="binomial")

Step 2: predict() function

predict(model2, data.frame(Mountains=0 + PriorWar=1 + GDPPC=1000, data=FL2003, type="response"))

predict(model2, data.frame(Mountains=0.5 + PriorWar=1 + GDPPC=1000, data=FL2003, type="response"))

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